

OOSTENDE PIER, Belgium

WMO: 064080

Lat: 51.239N

Lon: 2.917E

Elev: 49

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 90-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.8	27.5	18.3	13.8	25.2	22.1	16.7	29.8	46.4	49.2	42.5	50.4	8.2	70	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	78.3	68.7	74.5	66.7	72.0	65.3	69.6	75.8	67.8	73.1	66.2	71.3	8.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	101.2	72.5	65.6	94.6	71.2	64.0	89.3	69.8	33.7	76.0	32.1	73.4	30.9	71.4	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.8	34.6	30.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	20.7	73.4	5.9	3.8	16.4	76.1	13.0	78.3	9.7	80.5	5.4	83.2	(4)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	41.5	43.1	45.8	49.0	55.4	59.9	64.3	65.7	59.9	54.5	47.1	41.0
	DBStd	9.93	7.20	5.35	4.04	4.64	5.30	4.68	3.94	4.38	3.67	5.42	5.28	6.32
	HDD50	1101	267	199	138	70	9	0	0	0	0	23	112	283
	HDD65	4759	730	614	595	479	302	167	62	42	160	326	537	745
	CDD50	1947	2	4	8	42	176	297	443	488	297	162	25	3
	CDD65	131	0	0	0	0	4	14	41	65	7	0	0	0
	CDH74	362	0	0	0	0	25	47	90	190	10	0	0	0
	CDH80	81	0	0	0	0	8	10	13	50	0	0	0	0
Wind	WSAvg	13.4	14.8	16.3	13.9	13.3	12.9	12.9	12.5	12.1	11.2	12.5	14.3	14.7
Precipitation	PrecAvg	30.6	2.4	2.2	1.8	1.7	2.2	2.6	2.8	2.9	2.6	3.0	3.4	3.0
	PrecMax	37.9	4.7	3.8	3.7	3.1	5.6	5.8	6.3	6.2	6.6	5.9	6.9	7.2
	PrecMin	22.4	0.1	0.6	0.4	0.2	0.5	0.9	0.6	0.3	0.2	0.7	0.4	0.6
	PrecStd	4.2	1.1	1.0	0.9	0.8	1.1	1.2	1.3	1.8	1.7	1.3	1.4	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.7	58.1	62.3	69.2	75.7	80.2	82.1	86.9	75.0	70.1	60.9	55.5
		MCWB	52.0	51.8	51.6	55.8	63.2	64.6	68.1	70.7	64.7	61.7	55.6	53.3
	2%	DB	52.8	53.9	56.7	62.5	69.3	73.9	76.8	78.9	70.6	65.6	57.2	53.4
		MCWB	50.1	49.7	50.1	53.8	61.1	63.7	68.3	70.8	63.4	59.6	54.4	51.5
	5%	DB	51.2	51.6	53.4	57.8	65.5	69.5	73.2	75.1	67.7	63.2	55.2	51.0
		MCWB	48.7	48.1	48.7	52.2	59.5	62.4	66.0	68.6	61.9	58.9	52.4	49.0
	10%	DB	49.6	49.7	51.1	55.0	62.8	66.6	70.8	72.5	65.5	61.3	53.7	49.0
		MCWB	47.3	46.6	47.5	50.9	57.8	61.0	64.9	66.7	60.2	57.6	50.9	47.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.2	53.1	53.1	58.5	66.4	67.1	70.9	75.6	66.6	62.9	56.7	53.5
		MCDB	54.0	56.9	59.0	64.0	72.4	77.2	77.5	76.9	72.0	67.0	58.5	55.1
	2%	WB	50.5	50.7	51.0	55.1	62.6	64.8	68.5	70.8	64.6	61.0	54.7	51.5
		MCDB	52.3	53.3	54.9	61.1	65.9	72.1	74.1	77.3	68.9	64.5	56.4	53.2
	5%	WB	49.1	48.4	49.4	53.0	60.0	63.1	66.7	68.9	62.8	59.6	52.9	49.5
		MCDB	50.8	50.8	52.6	57.2	64.0	68.6	71.9	74.6	66.9	62.5	54.6	50.9
	10%	WB	47.5	46.9	48.1	51.3	57.9	61.4	65.1	67.3	61.0	58.2	51.4	47.3
		MCDB	49.2	49.4	50.8	54.6	62.4	65.9	70.1	72.2	64.6	60.9	53.4	48.8
Mean Daily Temperature Range	5% DB	MDBR	6.5	6.6	6.9	7.8	8.5	7.8	8.2	8.8	8.5	8.1	6.5	6.3
		MCDBR	8.1	9.2	12.1	13.7	13.4	13.4	12.8	14.2	13.0	10.6	7.7	7.6
		MCWBR	7.4	7.6	8.1	8.9	9.0	7.9	8.1	8.1	7.8	6.8	6.8	6.9
	5% WB	MCDBR	8.0	7.9	9.3	11.3	12.5	11.7	11.4	13.1	10.8	9.3	7.1	7.2
		MCWBR	7.7	7.4	6.9	7.9	9.7	7.4	8.0	8.5	7.8	7.2	7.0	7.0
Clear-Sky Solar Irradiance	taub	0.334	0.353	0.390	0.413	0.423	0.420	0.422	0.413	0.391	0.370	0.345	0.329	
	taud	2.370	2.357	2.257	2.226	2.251	2.280	2.284	2.326	2.373	2.407	2.381	2.348	
	Ebn at Noon	214	243	253	261	264	266	263	259	251	233	207	194	
	Edh at Noon	20	26	35	40	41	40	39	36	31	25	20	18	
All-Sky Solar Radiation	RadAvg	244	466	856	1398	1635	1770	1674	1429	1054	602	285	186	
	RadStd	21	85	107	128	121	140	169	119	70	52	29	23	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (72 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page